

SOLVED PRACTICE WORKBOOK

Landslides, Avalanches and GLOF Risk - Learner-v2 Refreshed

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing



Original deterministic study visual; labels are explanatory, not textual quotations.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

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Major learning parts and meaningful subtopics are listed in document order. Page numbers are generated from the final PDF layout; PDF bookmarks mirror the same hierarchy.

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BASIC MCQS / REMEDIATION

Q1. Which statement correctly identifies Landslide?

A. A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures. B. Rockfall is the detachment and rapid fall, bounce or roll of rock from a steep face or slope, distinct from a coherent slide or channelised debris flow. C. An avalanche is rapid downslope movement of snow, ice and entrained material produced by snowpack instability, slope and weather conditions; it is not a landslide label for every mountain failure. D. A debris flow is a rapid, channelised or slope-confined mixture of water, soil, rock and organic material; intense rainfall may trigger it but rainfall is not the moving mass.

Answer: A. Explanation: A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q2. Which option preserves the risk or institutional boundary of Landslide?

A. A debris flow is a rapid, channelised or slope-confined mixture of water, soil, rock and organic material; intense rainfall may trigger it but rainfall is not the moving mass. B. A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures. C. An avalanche is rapid downslope movement of snow, ice and entrained material produced by snowpack instability, slope and weather conditions; it is not a landslide label for every mountain failure. D. A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche.

Answer: B. Explanation: A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q3. Which statement uses Landslide without changing its hazard, mandate or status?

A. An avalanche is rapid downslope movement of snow, ice and entrained material produced by snowpack instability, slope and weather conditions; it is not a landslide label for every mountain failure. B. Susceptibility indicates where terrain is more likely to fail from relatively stable conditioning factors, whereas hazard adds probability, timing or intensity under triggers; neither alone equals risk. C. A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures. D. A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche.

Answer: C. Explanation: A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q4. Which option avoids the standard UPSC close-option trap about Landslide?

A. A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche. B. Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks. C. Susceptibility indicates where terrain is more likely to fail from relatively stable conditioning factors, whereas hazard adds probability, timing or intensity under triggers; neither alone equals risk. D. A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures.

Answer: D. Explanation: A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q5. Which statement correctly identifies Rockfall?

A. Rockfall is the detachment and rapid fall, bounce or roll of rock from a steep face or slope, distinct from a coherent slide or channelised debris flow. B. An avalanche is rapid downslope movement of snow, ice and entrained material produced by snowpack instability, slope and weather conditions; it is not a landslide label for every mountain failure. C. A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche. D. A debris flow is a rapid, channelised or slope-confined mixture of water, soil, rock and organic material; intense rainfall may trigger it but rainfall is not the moving mass.

Answer: A. Explanation: Rockfall is the detachment and rapid fall, bounce or roll of rock from a steep face or slope, distinct from a coherent slide or channelised debris flow. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q6. Which option preserves the risk or institutional boundary of Rockfall?

A. A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche. B. Rockfall is the detachment and rapid fall, bounce or roll of rock from a steep face or slope, distinct from a coherent slide or channelised debris flow. C. An avalanche is rapid downslope movement of snow, ice and entrained material produced by snowpack instability, slope and weather conditions; it is not a landslide label for every mountain failure. D. Susceptibility indicates where terrain is more likely to fail from relatively stable conditioning factors, whereas hazard adds probability, timing or intensity under triggers; neither alone equals risk.

Answer: B. Explanation: Rockfall is the detachment and rapid fall, bounce or roll of rock from a steep face or slope, distinct from a coherent slide or channelised debris flow. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q7. Which statement uses Rockfall without changing its hazard, mandate or status?

A. Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks. B. Susceptibility indicates where terrain is more likely to fail from relatively stable conditioning factors, whereas hazard adds probability, timing or intensity under triggers; neither alone equals risk. C. Rockfall is the detachment and rapid fall, bounce or roll of rock from a steep face or slope, distinct from a coherent slide or channelised debris flow. D. A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche.

Answer: C. Explanation: Rockfall is the detachment and rapid fall, bounce or roll of rock from a steep face or slope, distinct from a coherent slide or channelised debris flow. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q8. Which option avoids the standard UPSC close-option trap about Rockfall?

A. Susceptibility indicates where terrain is more likely to fail from relatively stable conditioning factors, whereas hazard adds probability, timing or intensity under triggers; neither alone equals risk. B. Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability. C. Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks. D. Rockfall is the detachment and rapid fall, bounce or roll of rock from a steep face or slope, distinct from a coherent slide or channelised debris flow.

Answer: D. Explanation: Rockfall is the detachment and rapid fall, bounce or roll of rock from a steep face or slope, distinct from a coherent slide or channelised debris flow. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q9. Which statement correctly identifies Debris flow?

A. A debris flow is a rapid, channelised or slope-confined mixture of water, soil, rock and organic material; intense rainfall may trigger it but rainfall is not the moving mass. B. Susceptibility indicates where terrain is more likely to fail from relatively stable conditioning factors, whereas hazard adds probability, timing or intensity under triggers; neither alone equals risk. C. A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche. D. An avalanche is rapid downslope movement of snow, ice and entrained material produced by snowpack instability, slope and weather conditions; it is not a landslide label for every mountain failure.

Answer: A. Explanation: A debris flow is a rapid, channelised or slope-confined mixture of water, soil, rock and organic material; intense rainfall may trigger it but rainfall is not the moving mass. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q10. Which option preserves the risk or institutional boundary of Debris flow?

A. A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche. B. A debris flow is a rapid, channelised or slope-confined mixture of water, soil, rock and organic material; intense rainfall may trigger it but rainfall is not the moving mass. C. Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks. D. Susceptibility indicates where terrain is more likely to fail from relatively stable conditioning factors, whereas hazard adds probability, timing or intensity under triggers; neither alone equals risk.

Answer: B. Explanation: A debris flow is a rapid, channelised or slope-confined mixture of water, soil, rock and organic material; intense rainfall may trigger it but rainfall is not the moving mass. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q11. Which statement uses Debris flow without changing its hazard, mandate or status?

A. Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks. B. Susceptibility indicates where terrain is more likely to fail from relatively stable conditioning factors, whereas hazard adds probability, timing or intensity under triggers; neither alone equals risk. C. A debris flow is a rapid, channelised or slope-confined mixture of water, soil, rock and organic material; intense rainfall may trigger it but rainfall is not the moving mass. D. Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability.

Answer: C. Explanation: A debris flow is a rapid, channelised or slope-confined mixture of water, soil, rock and organic material; intense rainfall may trigger it but rainfall is not the moving mass. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q12. Which option avoids the standard UPSC close-option trap about Debris flow?

A. Intense or prolonged rainfall, earthquakes, snow loading or warming, freeze-thaw, erosion, excavation and slope disturbance can trigger failure, but the relevant trigger differs among landslide, avalanche and GLOF mechanisms. B. Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability. C. Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks. D. A debris flow is a rapid, channelised or slope-confined mixture of water, soil, rock and organic material; intense rainfall may trigger it but rainfall is not the moving mass.

Answer: D. Explanation: A debris flow is a rapid, channelised or slope-confined mixture of water, soil, rock and organic material; intense rainfall may trigger it but rainfall is not the moving mass. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q13. Which statement correctly identifies Avalanche?

A. An avalanche is rapid downslope movement of snow, ice and entrained material produced by snowpack instability, slope and weather conditions; it is not a landslide label for every mountain failure. B. Susceptibility indicates where terrain is more likely to fail from relatively stable conditioning factors, whereas hazard adds probability, timing or intensity under triggers; neither alone equals risk. C. A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche. D. Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks.

Answer: A. Explanation: An avalanche is rapid downslope movement of snow, ice and entrained material produced by snowpack instability, slope and weather conditions; it is not a landslide label for every mountain failure. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q14. Which option preserves the risk or institutional boundary of Avalanche?

A. Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability. B. An avalanche is rapid downslope movement of snow, ice and entrained material produced by snowpack instability, slope and weather conditions; it is not a landslide label for every mountain failure. C. Susceptibility indicates where terrain is more likely to fail from relatively stable conditioning factors, whereas hazard adds probability, timing or intensity under triggers; neither alone equals risk. D. Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks.

Answer: B. Explanation: An avalanche is rapid downslope movement of snow, ice and entrained material produced by snowpack instability, slope and weather conditions; it is not a landslide label for every mountain failure. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q15. Which statement uses Avalanche without changing its hazard, mandate or status?

A. Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks. B. Intense or prolonged rainfall, earthquakes, snow loading or warming, freeze-thaw, erosion, excavation and slope disturbance can trigger failure, but the relevant trigger differs among landslide, avalanche and GLOF mechanisms. C. An avalanche is rapid downslope movement of snow, ice and entrained material produced by snowpack instability, slope and weather conditions; it is not a landslide label for every mountain failure. D. Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability.

Answer: C. Explanation: An avalanche is rapid downslope movement of snow, ice and entrained material produced by snowpack instability, slope and weather conditions; it is not a landslide label for every mountain failure. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q16. Which option avoids the standard UPSC close-option trap about Avalanche?

A. Intense or prolonged rainfall, earthquakes, snow loading or warming, freeze-thaw, erosion, excavation and slope disturbance can trigger failure, but the relevant trigger differs among landslide, avalanche and GLOF mechanisms. B. Earthquake or rain can trigger landslides that block rivers; lake or blockage failure can then generate flooding, while cloudburst, debris flow and slope failure may occur in one compound mountain emergency. C. Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability. D. An avalanche is rapid downslope movement of snow, ice and entrained material produced by snowpack instability, slope and weather conditions; it is not a landslide label for every mountain failure.

Answer: D. Explanation: An avalanche is rapid downslope movement of snow, ice and entrained material produced by snowpack instability, slope and weather conditions; it is not a landslide label for every mountain failure. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q17. Which statement correctly identifies GLOF?

A. A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche. B. Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability. C. Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks. D. Susceptibility indicates where terrain is more likely to fail from relatively stable conditioning factors, whereas hazard adds probability, timing or intensity under triggers; neither alone equals risk.

Answer: A. Explanation: A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q18. Which option preserves the risk or institutional boundary of GLOF?

A. Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks. B. A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche. C. Intense or prolonged rainfall, earthquakes, snow loading or warming, freeze-thaw, erosion, excavation and slope disturbance can trigger failure, but the relevant trigger differs among landslide, avalanche and GLOF mechanisms. D. Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability.

Answer: B. Explanation: A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q19. Which statement uses GLOF without changing its hazard, mandate or status?

A. Earthquake or rain can trigger landslides that block rivers; lake or blockage failure can then generate flooding, while cloudburst, debris flow and slope failure may occur in one compound mountain emergency. B. Intense or prolonged rainfall, earthquakes, snow loading or warming, freeze-thaw, erosion, excavation and slope disturbance can trigger failure, but the relevant trigger differs among landslide, avalanche and GLOF mechanisms. C. A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche. D. Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability.

Answer: C. Explanation: A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q20. Which option avoids the standard UPSC close-option trap about GLOF?

A. Intense or prolonged rainfall, earthquakes, snow loading or warming, freeze-thaw, erosion, excavation and slope disturbance can trigger failure, but the relevant trigger differs among landslide, avalanche and GLOF mechanisms. B. Earthquake or rain can trigger landslides that block rivers; lake or blockage failure can then generate flooding, while cloudburst, debris flow and slope failure may occur in one compound mountain emergency. C. GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work. D. A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche.

Answer: D. Explanation: A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q21. Which statement correctly identifies Susceptibility versus hazard?

A. Susceptibility indicates where terrain is more likely to fail from relatively stable conditioning factors, whereas hazard adds probability, timing or intensity under triggers; neither alone equals risk. B. Intense or prolonged rainfall, earthquakes, snow loading or warming, freeze-thaw, erosion, excavation and slope disturbance can trigger failure, but the relevant trigger differs among landslide, avalanche and GLOF mechanisms. C. Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks. D. Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability.

Answer: A. Explanation: Susceptibility indicates where terrain is more likely to fail from relatively stable conditioning factors, whereas hazard adds probability, timing or intensity under triggers; neither alone equals risk. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q22. Which option preserves the risk or institutional boundary of Susceptibility versus hazard?

A. Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability. B. Susceptibility indicates where terrain is more likely to fail from relatively stable conditioning factors, whereas hazard adds probability, timing or intensity under triggers; neither alone equals risk. C. Intense or prolonged rainfall, earthquakes, snow loading or warming, freeze-thaw, erosion, excavation and slope disturbance can trigger failure, but the relevant trigger differs among landslide, avalanche and GLOF mechanisms. D. Earthquake or rain can trigger landslides that block rivers; lake or blockage failure can then generate flooding, while cloudburst, debris flow and slope failure may occur in one compound mountain emergency.

Answer: B. Explanation: Susceptibility indicates where terrain is more likely to fail from relatively stable conditioning factors, whereas hazard adds probability, timing or intensity under triggers; neither alone equals risk. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q23. Which statement uses Susceptibility versus hazard without changing its hazard, mandate or status?

A. Earthquake or rain can trigger landslides that block rivers; lake or blockage failure can then generate flooding, while cloudburst, debris flow and slope failure may occur in one compound mountain emergency. B. Intense or prolonged rainfall, earthquakes, snow loading or warming, freeze-thaw, erosion, excavation and slope disturbance can trigger failure, but the relevant trigger differs among landslide, avalanche and GLOF mechanisms. C. Susceptibility indicates where terrain is more likely to fail from relatively stable conditioning factors, whereas hazard adds probability, timing or intensity under triggers; neither alone equals risk. D. GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work.

Answer: C. Explanation: Susceptibility indicates where terrain is more likely to fail from relatively stable conditioning factors, whereas hazard adds probability, timing or intensity under triggers; neither alone equals risk. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q24. Which option avoids the standard UPSC close-option trap about Susceptibility versus hazard?

A. DGRE under DRDO is the hazard-specific institutional anchor for snow and avalanche observation, assessment and warnings; its bulletin is not a GSI landslide forecast. B. Earthquake or rain can trigger landslides that block rivers; lake or blockage failure can then generate flooding, while cloudburst, debris flow and slope failure may occur in one compound mountain emergency. C. GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work. D. Susceptibility indicates where terrain is more likely to fail from relatively stable conditioning factors, whereas hazard adds probability, timing or intensity under triggers; neither alone equals risk.

Answer: D. Explanation: Susceptibility indicates where terrain is more likely to fail from relatively stable conditioning factors, whereas hazard adds probability, timing or intensity under triggers; neither alone equals risk. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q25. Which statement correctly identifies Risk?

A. Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks. B. Earthquake or rain can trigger landslides that block rivers; lake or blockage failure can then generate flooding, while cloudburst, debris flow and slope failure may occur in one compound mountain emergency. C. Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability. D. Intense or prolonged rainfall, earthquakes, snow loading or warming, freeze-thaw, erosion, excavation and slope disturbance can trigger failure, but the relevant trigger differs among landslide, avalanche and GLOF mechanisms.

Answer: A. Explanation: Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q26. Which option preserves the risk or institutional boundary of Risk?

A. Intense or prolonged rainfall, earthquakes, snow loading or warming, freeze-thaw, erosion, excavation and slope disturbance can trigger failure, but the relevant trigger differs among landslide, avalanche and GLOF mechanisms. B. Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks. C. Earthquake or rain can trigger landslides that block rivers; lake or blockage failure can then generate flooding, while cloudburst, debris flow and slope failure may occur in one compound mountain emergency. D. GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work.

Answer: B. Explanation: Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q27. Which statement uses Risk without changing its hazard, mandate or status?

A. Earthquake or rain can trigger landslides that block rivers; lake or blockage failure can then generate flooding, while cloudburst, debris flow and slope failure may occur in one compound mountain emergency. B. DGRE under DRDO is the hazard-specific institutional anchor for snow and avalanche observation, assessment and warnings; its bulletin is not a GSI landslide forecast. C. Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks. D. GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work.

Answer: C. Explanation: Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q28. Which option avoids the standard UPSC close-option trap about Risk?

A. DGRE under DRDO is the hazard-specific institutional anchor for snow and avalanche observation, assessment and warnings; its bulletin is not a GSI landslide forecast. B. GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work. C. NRSC and ISRO provide Earth-observation inventories and mapping support for landslides and glacial or water-body monitoring; an image or atlas is decision support, not proof of safe slopes or warned communities. D. Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks.

Answer: D. Explanation: Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q29. Which statement correctly identifies Conditioning factors?

A. Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability. B. GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work. C. Earthquake or rain can trigger landslides that block rivers; lake or blockage failure can then generate flooding, while cloudburst, debris flow and slope failure may occur in one compound mountain emergency. D. Intense or prolonged rainfall, earthquakes, snow loading or warming, freeze-thaw, erosion, excavation and slope disturbance can trigger failure, but the relevant trigger differs among landslide, avalanche and GLOF mechanisms.

Answer: A. Explanation: Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q30. Which option preserves the risk or institutional boundary of Conditioning factors?

A. GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work. B. Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability. C. DGRE under DRDO is the hazard-specific institutional anchor for snow and avalanche observation, assessment and warnings; its bulletin is not a GSI landslide forecast. D. Earthquake or rain can trigger landslides that block rivers; lake or blockage failure can then generate flooding, while cloudburst, debris flow and slope failure may occur in one compound mountain emergency.

Answer: B. Explanation: Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q31. Which statement uses Conditioning factors without changing its hazard, mandate or status?

A. DGRE under DRDO is the hazard-specific institutional anchor for snow and avalanche observation, assessment and warnings; its bulletin is not a GSI landslide forecast. B. NRSC and ISRO provide Earth-observation inventories and mapping support for landslides and glacial or water-body monitoring; an image or atlas is decision support, not proof of safe slopes or warned communities. C. Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability. D. GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work.

Answer: C. Explanation: Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q32. Which option avoids the standard UPSC close-option trap about Conditioning factors?

A. NRSC and ISRO provide Earth-observation inventories and mapping support for landslides and glacial or water-body monitoring; an image or atlas is decision support, not proof of safe slopes or warned communities. B. DGRE under DRDO is the hazard-specific institutional anchor for snow and avalanche observation, assessment and warnings; its bulletin is not a GSI landslide forecast. C. Glacial-lake risk screening requires repeated remote sensing, lake and dam-character assessment, upstream slope or ice-change observation, field validation where feasible and downstream exposure mapping. D. Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability.

Answer: D. Explanation: Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q33. Which statement correctly identifies Triggers?

A. Intense or prolonged rainfall, earthquakes, snow loading or warming, freeze-thaw, erosion, excavation and slope disturbance can trigger failure, but the relevant trigger differs among landslide, avalanche and GLOF mechanisms. B. GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work. C. Earthquake or rain can trigger landslides that block rivers; lake or blockage failure can then generate flooding, while cloudburst, debris flow and slope failure may occur in one compound mountain emergency. D. DGRE under DRDO is the hazard-specific institutional anchor for snow and avalanche observation, assessment and warnings; its bulletin is not a GSI landslide forecast.

Answer: A. Explanation: Intense or prolonged rainfall, earthquakes, snow loading or warming, freeze-thaw, erosion, excavation and slope disturbance can trigger failure, but the relevant trigger differs among landslide, avalanche and GLOF mechanisms. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q34. Which option preserves the risk or institutional boundary of Triggers?

A. GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work. B. Intense or prolonged rainfall, earthquakes, snow loading or warming, freeze-thaw, erosion, excavation and slope disturbance can trigger failure, but the relevant trigger differs among landslide, avalanche and GLOF mechanisms. C. NRSC and ISRO provide Earth-observation inventories and mapping support for landslides and glacial or water-body monitoring; an image or atlas is decision support, not proof of safe slopes or warned communities. D. DGRE under DRDO is the hazard-specific institutional anchor for snow and avalanche observation, assessment and warnings; its bulletin is not a GSI landslide forecast.

Answer: B. Explanation: Intense or prolonged rainfall, earthquakes, snow loading or warming, freeze-thaw, erosion, excavation and slope disturbance can trigger failure, but the relevant trigger differs among landslide, avalanche and GLOF mechanisms. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q35. Which statement uses Triggers without changing its hazard, mandate or status?

A. DGRE under DRDO is the hazard-specific institutional anchor for snow and avalanche observation, assessment and warnings; its bulletin is not a GSI landslide forecast. B. Glacial-lake risk screening requires repeated remote sensing, lake and dam-character assessment, upstream slope or ice-change observation, field validation where feasible and downstream exposure mapping. C. Intense or prolonged rainfall, earthquakes, snow loading or warming, freeze-thaw, erosion, excavation and slope disturbance can trigger failure, but the relevant trigger differs among landslide, avalanche and GLOF mechanisms. D. NRSC and ISRO provide Earth-observation inventories and mapping support for landslides and glacial or water-body monitoring; an image or atlas is decision support, not proof of safe slopes or warned communities.

Answer: C. Explanation: Intense or prolonged rainfall, earthquakes, snow loading or warming, freeze-thaw, erosion, excavation and slope disturbance can trigger failure, but the relevant trigger differs among landslide, avalanche and GLOF mechanisms. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q36. Which option avoids the standard UPSC close-option trap about Triggers?

A. Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan. B. Glacial-lake risk screening requires repeated remote sensing, lake and dam-character assessment, upstream slope or ice-change observation, field validation where feasible and downstream exposure mapping. C. NRSC and ISRO provide Earth-observation inventories and mapping support for landslides and glacial or water-body monitoring; an image or atlas is decision support, not proof of safe slopes or warned communities. D. Intense or prolonged rainfall, earthquakes, snow loading or warming, freeze-thaw, erosion, excavation and slope disturbance can trigger failure, but the relevant trigger differs among landslide, avalanche and GLOF mechanisms.

Answer: D. Explanation: Intense or prolonged rainfall, earthquakes, snow loading or warming, freeze-thaw, erosion, excavation and slope disturbance can trigger failure, but the relevant trigger differs among landslide, avalanche and GLOF mechanisms. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q37. Which statement correctly identifies Cascading risk?

A. Earthquake or rain can trigger landslides that block rivers; lake or blockage failure can then generate flooding, while cloudburst, debris flow and slope failure may occur in one compound mountain emergency. B. DGRE under DRDO is the hazard-specific institutional anchor for snow and avalanche observation, assessment and warnings; its bulletin is not a GSI landslide forecast. C. NRSC and ISRO provide Earth-observation inventories and mapping support for landslides and glacial or water-body monitoring; an image or atlas is decision support, not proof of safe slopes or warned communities. D. GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work.

Answer: A. Explanation: Earthquake or rain can trigger landslides that block rivers; lake or blockage failure can then generate flooding, while cloudburst, debris flow and slope failure may occur in one compound mountain emergency. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q38. Which option preserves the risk or institutional boundary of Cascading risk?

A. Glacial-lake risk screening requires repeated remote sensing, lake and dam-character assessment, upstream slope or ice-change observation, field validation where feasible and downstream exposure mapping. B. Earthquake or rain can trigger landslides that block rivers; lake or blockage failure can then generate flooding, while cloudburst, debris flow and slope failure may occur in one compound mountain emergency. C. DGRE under DRDO is the hazard-specific institutional anchor for snow and avalanche observation, assessment and warnings; its bulletin is not a GSI landslide forecast. D. NRSC and ISRO provide Earth-observation inventories and mapping support for landslides and glacial or water-body monitoring; an image or atlas is decision support, not proof of safe slopes or warned communities.

Answer: B. Explanation: Earthquake or rain can trigger landslides that block rivers; lake or blockage failure can then generate flooding, while cloudburst, debris flow and slope failure may occur in one compound mountain emergency. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q39. Which statement uses Cascading risk without changing its hazard, mandate or status?

A. NRSC and ISRO provide Earth-observation inventories and mapping support for landslides and glacial or water-body monitoring; an image or atlas is decision support, not proof of safe slopes or warned communities. B. Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan. C. Earthquake or rain can trigger landslides that block rivers; lake or blockage failure can then generate flooding, while cloudburst, debris flow and slope failure may occur in one compound mountain emergency. D. Glacial-lake risk screening requires repeated remote sensing, lake and dam-character assessment, upstream slope or ice-change observation, field validation where feasible and downstream exposure mapping.

Answer: C. Explanation: Earthquake or rain can trigger landslides that block rivers; lake or blockage failure can then generate flooding, while cloudburst, debris flow and slope failure may occur in one compound mountain emergency. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q40. Which option avoids the standard UPSC close-option trap about Cascading risk?

A. Glacial-lake risk screening requires repeated remote sensing, lake and dam-character assessment, upstream slope or ice-change observation, field validation where feasible and downstream exposure mapping. B. Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan. C. Surface and subsurface drainage, controlled excavation, retaining or protective works, bioengineering and maintenance may reduce selected instability, but engineering must be site-specific and cannot guarantee every slope. D. Earthquake or rain can trigger landslides that block rivers; lake or blockage failure can then generate flooding, while cloudburst, debris flow and slope failure may occur in one compound mountain emergency.

Answer: D. Explanation: Earthquake or rain can trigger landslides that block rivers; lake or blockage failure can then generate flooding, while cloudburst, debris flow and slope failure may occur in one compound mountain emergency. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q41. Which statement correctly identifies GSI role?

A. GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work. B. DGRE under DRDO is the hazard-specific institutional anchor for snow and avalanche observation, assessment and warnings; its bulletin is not a GSI landslide forecast. C. Glacial-lake risk screening requires repeated remote sensing, lake and dam-character assessment, upstream slope or ice-change observation, field validation where feasible and downstream exposure mapping. D. NRSC and ISRO provide Earth-observation inventories and mapping support for landslides and glacial or water-body monitoring; an image or atlas is decision support, not proof of safe slopes or warned communities.

Answer: A. Explanation: GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q42. Which option preserves the risk or institutional boundary of GSI role?

A. NRSC and ISRO provide Earth-observation inventories and mapping support for landslides and glacial or water-body monitoring; an image or atlas is decision support, not proof of safe slopes or warned communities. B. GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work. C. Glacial-lake risk screening requires repeated remote sensing, lake and dam-character assessment, upstream slope or ice-change observation, field validation where feasible and downstream exposure mapping. D. Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan.

Answer: B. Explanation: GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q43. Which statement uses GSI role without changing its hazard, mandate or status?

A. Surface and subsurface drainage, controlled excavation, retaining or protective works, bioengineering and maintenance may reduce selected instability, but engineering must be site-specific and cannot guarantee every slope. B. Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan. C. GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work. D. Glacial-lake risk screening requires repeated remote sensing, lake and dam-character assessment, upstream slope or ice-change observation, field validation where feasible and downstream exposure mapping.

Answer: C. Explanation: GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q44. Which option avoids the standard UPSC close-option trap about GSI role?

A. Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services. B. Surface and subsurface drainage, controlled excavation, retaining or protective works, bioengineering and maintenance may reduce selected instability, but engineering must be site-specific and cannot guarantee every slope. C. Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan. D. GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work.

Answer: D. Explanation: GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q45. Which statement correctly identifies DGRE role?

A. DGRE under DRDO is the hazard-specific institutional anchor for snow and avalanche observation, assessment and warnings; its bulletin is not a GSI landslide forecast. B. NRSC and ISRO provide Earth-observation inventories and mapping support for landslides and glacial or water-body monitoring; an image or atlas is decision support, not proof of safe slopes or warned communities. C. Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan. D. Glacial-lake risk screening requires repeated remote sensing, lake and dam-character assessment, upstream slope or ice-change observation, field validation where feasible and downstream exposure mapping.

Answer: A. Explanation: DGRE under DRDO is the hazard-specific institutional anchor for snow and avalanche observation, assessment and warnings; its bulletin is not a GSI landslide forecast. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q46. Which option preserves the risk or institutional boundary of DGRE role?

A. Glacial-lake risk screening requires repeated remote sensing, lake and dam-character assessment, upstream slope or ice-change observation, field validation where feasible and downstream exposure mapping. B. DGRE under DRDO is the hazard-specific institutional anchor for snow and avalanche observation, assessment and warnings; its bulletin is not a GSI landslide forecast. C. Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan. D. Surface and subsurface drainage, controlled excavation, retaining or protective works, bioengineering and maintenance may reduce selected instability, but engineering must be site-specific and cannot guarantee every slope.

Answer: B. Explanation: DGRE under DRDO is the hazard-specific institutional anchor for snow and avalanche observation, assessment and warnings; its bulletin is not a GSI landslide forecast. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q47. Which statement uses DGRE role without changing its hazard, mandate or status?

A. Surface and subsurface drainage, controlled excavation, retaining or protective works, bioengineering and maintenance may reduce selected instability, but engineering must be site-specific and cannot guarantee every slope. B. Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan. C. DGRE under DRDO is the hazard-specific institutional anchor for snow and avalanche observation, assessment and warnings; its bulletin is not a GSI landslide forecast. D. Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services.

Answer: C. Explanation: DGRE under DRDO is the hazard-specific institutional anchor for snow and avalanche observation, assessment and warnings; its bulletin is not a GSI landslide forecast. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q48. Which option avoids the standard UPSC close-option trap about DGRE role?

A. Surface and subsurface drainage, controlled excavation, retaining or protective works, bioengineering and maintenance may reduce selected instability, but engineering must be site-specific and cannot guarantee every slope. B. Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback. C. Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services. D. DGRE under DRDO is the hazard-specific institutional anchor for snow and avalanche observation, assessment and warnings; its bulletin is not a GSI landslide forecast.

Answer: D. Explanation: DGRE under DRDO is the hazard-specific institutional anchor for snow and avalanche observation, assessment and warnings; its bulletin is not a GSI landslide forecast. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q49. Which statement correctly identifies NRSC-ISRO role?

A. NRSC and ISRO provide Earth-observation inventories and mapping support for landslides and glacial or water-body monitoring; an image or atlas is decision support, not proof of safe slopes or warned communities. B. Glacial-lake risk screening requires repeated remote sensing, lake and dam-character assessment, upstream slope or ice-change observation, field validation where feasible and downstream exposure mapping. C. Surface and subsurface drainage, controlled excavation, retaining or protective works, bioengineering and maintenance may reduce selected instability, but engineering must be site-specific and cannot guarantee every slope. D. Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan.

Answer: A. Explanation: NRSC and ISRO provide Earth-observation inventories and mapping support for landslides and glacial or water-body monitoring; an image or atlas is decision support, not proof of safe slopes or warned communities. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q50. Which option preserves the risk or institutional boundary of NRSC-ISRO role?

A. Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services. B. NRSC and ISRO provide Earth-observation inventories and mapping support for landslides and glacial or water-body monitoring; an image or atlas is decision support, not proof of safe slopes or warned communities. C. Surface and subsurface drainage, controlled excavation, retaining or protective works, bioengineering and maintenance may reduce selected instability, but engineering must be site-specific and cannot guarantee every slope. D. Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan.

Answer: B. Explanation: NRSC and ISRO provide Earth-observation inventories and mapping support for landslides and glacial or water-body monitoring; an image or atlas is decision support, not proof of safe slopes or warned communities. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q51. Which statement uses NRSC-ISRO role without changing its hazard, mandate or status?

A. Surface and subsurface drainage, controlled excavation, retaining or protective works, bioengineering and maintenance may reduce selected instability, but engineering must be site-specific and cannot guarantee every slope. B. Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback. C. NRSC and ISRO provide Earth-observation inventories and mapping support for landslides and glacial or water-body monitoring; an image or atlas is decision support, not proof of safe slopes or warned communities. D. Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services.

Answer: C. Explanation: NRSC and ISRO provide Earth-observation inventories and mapping support for landslides and glacial or water-body monitoring; an image or atlas is decision support, not proof of safe slopes or warned communities. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q52. Which option avoids the standard UPSC close-option trap about NRSC-ISRO role?

A. Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback. B. Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services. C. GSI, DGRE, NDMA, CWC, NRSC/ISRO, State and district authorities, road agencies, hydropower operators and local communities hold different information and action roles that must be pre-agreed. D. NRSC and ISRO provide Earth-observation inventories and mapping support for landslides and glacial or water-body monitoring; an image or atlas is decision support, not proof of safe slopes or warned communities.

Answer: D. Explanation: NRSC and ISRO provide Earth-observation inventories and mapping support for landslides and glacial or water-body monitoring; an image or atlas is decision support, not proof of safe slopes or warned communities. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q53. Which statement correctly identifies Glacial-lake monitoring?

A. Glacial-lake risk screening requires repeated remote sensing, lake and dam-character assessment, upstream slope or ice-change observation, field validation where feasible and downstream exposure mapping. B. Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan. C. Surface and subsurface drainage, controlled excavation, retaining or protective works, bioengineering and maintenance may reduce selected instability, but engineering must be site-specific and cannot guarantee every slope. D. Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services.

Answer: A. Explanation: Glacial-lake risk screening requires repeated remote sensing, lake and dam-character assessment, upstream slope or ice-change observation, field validation where feasible and downstream exposure mapping. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q54. Which option preserves the risk or institutional boundary of Glacial-lake monitoring?

A. Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services. B. Glacial-lake risk screening requires repeated remote sensing, lake and dam-character assessment, upstream slope or ice-change observation, field validation where feasible and downstream exposure mapping. C. Surface and subsurface drainage, controlled excavation, retaining or protective works, bioengineering and maintenance may reduce selected instability, but engineering must be site-specific and cannot guarantee every slope. D. Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback.

Answer: B. Explanation: Glacial-lake risk screening requires repeated remote sensing, lake and dam-character assessment, upstream slope or ice-change observation, field validation where feasible and downstream exposure mapping. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q55. Which statement uses Glacial-lake monitoring without changing its hazard, mandate or status?

A. Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services. B. GSI, DGRE, NDMA, CWC, NRSC/ISRO, State and district authorities, road agencies, hydropower operators and local communities hold different information and action roles that must be pre-agreed. C. Glacial-lake risk screening requires repeated remote sensing, lake and dam-character assessment, upstream slope or ice-change observation, field validation where feasible and downstream exposure mapping. D. Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback.

Answer: C. Explanation: Glacial-lake risk screening requires repeated remote sensing, lake and dam-character assessment, upstream slope or ice-change observation, field validation where feasible and downstream exposure mapping. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q56. Which option avoids the standard UPSC close-option trap about Glacial-lake monitoring?

A. Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback. B. GSI, DGRE, NDMA, CWC, NRSC/ISRO, State and district authorities, road agencies, hydropower operators and local communities hold different information and action roles that must be pre-agreed. C. A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence. D. Glacial-lake risk screening requires repeated remote sensing, lake and dam-character assessment, upstream slope or ice-change observation, field validation where feasible and downstream exposure mapping.

Answer: D. Explanation: Glacial-lake risk screening requires repeated remote sensing, lake and dam-character assessment, upstream slope or ice-change observation, field validation where feasible and downstream exposure mapping. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q57. Which statement correctly identifies Zoning and land use?

A. Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan. B. Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback. C. Surface and subsurface drainage, controlled excavation, retaining or protective works, bioengineering and maintenance may reduce selected instability, but engineering must be site-specific and cannot guarantee every slope. D. Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services.

Answer: A. Explanation: Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q58. Which option preserves the risk or institutional boundary of Zoning and land use?

A. Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback. B. Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan. C. Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services. D. GSI, DGRE, NDMA, CWC, NRSC/ISRO, State and district authorities, road agencies, hydropower operators and local communities hold different information and action roles that must be pre-agreed.

Answer: B. Explanation: Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q59. Which statement uses Zoning and land use without changing its hazard, mandate or status?

A. Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback. B. GSI, DGRE, NDMA, CWC, NRSC/ISRO, State and district authorities, road agencies, hydropower operators and local communities hold different information and action roles that must be pre-agreed. C. Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan. D. A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence.

Answer: C. Explanation: Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q60. Which option avoids the standard UPSC close-option trap about Zoning and land use?

A. A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures. B. A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence. C. GSI, DGRE, NDMA, CWC, NRSC/ISRO, State and district authorities, road agencies, hydropower operators and local communities hold different information and action roles that must be pre-agreed. D. Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan.

Answer: D. Explanation: Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q61. Which statement correctly identifies Slope and drainage measures?

A. Surface and subsurface drainage, controlled excavation, retaining or protective works, bioengineering and maintenance may reduce selected instability, but engineering must be site-specific and cannot guarantee every slope. B. Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services. C. Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback. D. GSI, DGRE, NDMA, CWC, NRSC/ISRO, State and district authorities, road agencies, hydropower operators and local communities hold different information and action roles that must be pre-agreed.

Answer: A. Explanation: Surface and subsurface drainage, controlled excavation, retaining or protective works, bioengineering and maintenance may reduce selected instability, but engineering must be site-specific and cannot guarantee every slope. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q62. Which option preserves the risk or institutional boundary of Slope and drainage measures?

A. A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence. B. Surface and subsurface drainage, controlled excavation, retaining or protective works, bioengineering and maintenance may reduce selected instability, but engineering must be site-specific and cannot guarantee every slope. C. GSI, DGRE, NDMA, CWC, NRSC/ISRO, State and district authorities, road agencies, hydropower operators and local communities hold different information and action roles that must be pre-agreed. D. Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback.

Answer: B. Explanation: Surface and subsurface drainage, controlled excavation, retaining or protective works, bioengineering and maintenance may reduce selected instability, but engineering must be site-specific and cannot guarantee every slope. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q63. Which statement uses Slope and drainage measures without changing its hazard, mandate or status?

A. A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures. B. A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence. C. Surface and subsurface drainage, controlled excavation, retaining or protective works, bioengineering and maintenance may reduce selected instability, but engineering must be site-specific and cannot guarantee every slope. D. GSI, DGRE, NDMA, CWC, NRSC/ISRO, State and district authorities, road agencies, hydropower operators and local communities hold different information and action roles that must be pre-agreed.

Answer: C. Explanation: Surface and subsurface drainage, controlled excavation, retaining or protective works, bioengineering and maintenance may reduce selected instability, but engineering must be site-specific and cannot guarantee every slope. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q64. Which option avoids the standard UPSC close-option trap about Slope and drainage measures?

A. A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures. B. Rockfall is the detachment and rapid fall, bounce or roll of rock from a steep face or slope, distinct from a coherent slide or channelised debris flow. C. A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence. D. Surface and subsurface drainage, controlled excavation, retaining or protective works, bioengineering and maintenance may reduce selected instability, but engineering must be site-specific and cannot guarantee every slope.

Answer: D. Explanation: Surface and subsurface drainage, controlled excavation, retaining or protective works, bioengineering and maintenance may reduce selected instability, but engineering must be site-specific and cannot guarantee every slope. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q65. Which statement correctly identifies Route and settlement planning?

A. Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services. B. A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence. C. GSI, DGRE, NDMA, CWC, NRSC/ISRO, State and district authorities, road agencies, hydropower operators and local communities hold different information and action roles that must be pre-agreed. D. Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback.

Answer: A. Explanation: Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q66. Which option preserves the risk or institutional boundary of Route and settlement planning?

A. GSI, DGRE, NDMA, CWC, NRSC/ISRO, State and district authorities, road agencies, hydropower operators and local communities hold different information and action roles that must be pre-agreed. B. Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services. C. A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence. D. A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures.

Answer: B. Explanation: Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q67. Which statement uses Route and settlement planning without changing its hazard, mandate or status?

A. A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures. B. A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence. C. Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services. D. Rockfall is the detachment and rapid fall, bounce or roll of rock from a steep face or slope, distinct from a coherent slide or channelised debris flow.

Answer: C. Explanation: Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q68. Which option avoids the standard UPSC close-option trap about Route and settlement planning?

A. Rockfall is the detachment and rapid fall, bounce or roll of rock from a steep face or slope, distinct from a coherent slide or channelised debris flow. B. A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures. C. A debris flow is a rapid, channelised or slope-confined mixture of water, soil, rock and organic material; intense rainfall may trigger it but rainfall is not the moving mass. D. Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services.

Answer: D. Explanation: Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q69. Which statement correctly identifies Warning and evacuation?

A. Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback. B. A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures. C. A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence. D. GSI, DGRE, NDMA, CWC, NRSC/ISRO, State and district authorities, road agencies, hydropower operators and local communities hold different information and action roles that must be pre-agreed.

Answer: A. Explanation: Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q70. Which option preserves the risk or institutional boundary of Warning and evacuation?

A. Rockfall is the detachment and rapid fall, bounce or roll of rock from a steep face or slope, distinct from a coherent slide or channelised debris flow. B. Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback. C. A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence. D. A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures.

Answer: B. Explanation: Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q71. Which statement uses Warning and evacuation without changing its hazard, mandate or status?

A. Rockfall is the detachment and rapid fall, bounce or roll of rock from a steep face or slope, distinct from a coherent slide or channelised debris flow. B. A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures. C. Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback. D. A debris flow is a rapid, channelised or slope-confined mixture of water, soil, rock and organic material; intense rainfall may trigger it but rainfall is not the moving mass.

Answer: C. Explanation: Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and

feedback. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q72. Which option avoids the standard UPSC close-option trap about Warning and evacuation?

A. A debris flow is a rapid, channelised or slope-confined mixture of water, soil, rock and organic material; intense rainfall may trigger it but rainfall is not the moving mass. B. An avalanche is rapid downslope movement of snow, ice and entrained material produced by snowpack instability, slope and weather conditions; it is not a landslide label for every mountain failure. C. Rockfall is the detachment and rapid fall, bounce or roll of rock from a steep face or slope, distinct from a coherent slide or channelised debris flow. D. Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback.

Answer: D. Explanation: Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q73. Which statement correctly identifies Institutional coordination?

A. GSI, DGRE, NDMA, CWC, NRSC/ISRO, State and district authorities, road agencies, hydropower operators and local communities hold different information and action roles that must be pre-agreed. B. A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures. C. Rockfall is the detachment and rapid fall, bounce or roll of rock from a steep face or slope, distinct from a coherent slide or channelised debris flow. D. A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence.

Answer: A. Explanation: GSI, DGRE, NDMA, CWC, NRSC/ISRO, State and district authorities, road agencies, hydropower operators and local communities hold different information and action roles that must be pre-agreed. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q74. Which option preserves the risk or institutional boundary of Institutional coordination?

A. Rockfall is the detachment and rapid fall, bounce or roll of rock from a steep face or slope, distinct from a coherent slide or channelised debris flow. B. GSI, DGRE, NDMA, CWC, NRSC/ISRO, State and district authorities, road agencies, hydropower operators and local communities hold different information and action roles that must be pre-agreed. C. A debris flow is a rapid, channelised or slope-confined mixture of water, soil, rock and organic material; intense rainfall may trigger it but rainfall is not the moving mass. D. A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures.

Answer: B. Explanation: GSI, DGRE, NDMA, CWC, NRSC/ISRO, State and district authorities, road agencies, hydropower operators and local communities hold different information and action roles that must be pre-agreed. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q75. Which statement uses Institutional coordination without changing its hazard, mandate or status?

A. Rockfall is the detachment and rapid fall, bounce or roll of rock from a steep face or slope, distinct from a coherent slide or channelised debris flow. B. An avalanche is rapid downslope movement of snow, ice and entrained material produced by snowpack instability, slope and weather conditions; it is not a landslide label for every mountain failure. C. GSI, DGRE, NDMA, CWC, NRSC/ISRO, State and district authorities, road agencies, hydropower operators and local communities hold different information and action roles that must be pre-agreed. D. A debris flow is a rapid, channelised or slope-confined mixture of water, soil, rock and organic material; intense rainfall may trigger it but rainfall is not the moving mass.

Answer: C. Explanation: GSI, DGRE, NDMA, CWC, NRSC/ISRO, State and district authorities, road agencies, hydropower operators and local communities hold different information and action roles that must be

pre-agreed. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q76. Which option avoids the standard UPSC close-option trap about Institutional coordination?

A. An avalanche is rapid downslope movement of snow, ice and entrained material produced by snowpack instability, slope and weather conditions; it is not a landslide label for every mountain failure. B. A debris flow is a rapid, channelised or slope-confined mixture of water, soil, rock and organic material; intense rainfall may trigger it but rainfall is not the moving mass. C. A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche. D. GSI, DGRE, NDMA, CWC, NRSC/ISRO, State and district authorities, road agencies, hydropower operators and local communities hold different information and action roles that must be pre-agreed.

Answer: D. Explanation: GSI, DGRE, NDMA, CWC, NRSC/ISRO, State and district authorities, road agencies, hydropower operators and local communities hold different information and action roles that must be pre-agreed. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q77. Which statement correctly identifies Technology-outcome firewall?

A. A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence. B. A debris flow is a rapid, channelised or slope-confined mixture of water, soil, rock and organic material; intense rainfall may trigger it but rainfall is not the moving mass. C. Rockfall is the detachment and rapid fall, bounce or roll of rock from a steep face or slope, distinct from a coherent slide or channelised debris flow. D. A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures.

Answer: A. Explanation: A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q78. Which option preserves the risk or institutional boundary of Technology-outcome firewall?

A. A debris flow is a rapid, channelised or slope-confined mixture of water, soil, rock and organic material; intense rainfall may trigger it but rainfall is not the moving mass. B. A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence. C. Rockfall is the detachment and rapid fall, bounce or roll of rock from a steep face or slope, distinct from a coherent slide or channelised debris flow. D. An avalanche is rapid downslope movement of snow, ice and entrained material produced by snowpack instability, slope and weather conditions; it is not a landslide label for every mountain failure.

Answer: B. Explanation: A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q79. Which statement uses Technology-outcome firewall without changing its hazard, mandate or status?

A. A debris flow is a rapid, channelised or slope-confined mixture of water, soil, rock and organic material; intense rainfall may trigger it but rainfall is not the moving mass. B. An avalanche is rapid downslope movement of snow, ice and entrained material produced by snowpack instability, slope and weather conditions; it is not a landslide label for every mountain failure. C. A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence. D. A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche.

Answer: C. Explanation: A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q80. Which option avoids the standard UPSC close-option trap about Technology-outcome firewall?

A. An avalanche is rapid downslope movement of snow, ice and entrained material produced by snowpack instability, slope and weather conditions; it is not a landslide label for every mountain failure. B. Susceptibility indicates where terrain is more likely to fail from relatively stable conditioning factors, whereas hazard adds probability, timing or intensity under triggers; neither alone equals risk. C. A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche. D. A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence.

Answer: D. Explanation: A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

PYQS AND ANSWER PRACTICE

AUDITED TOPIC-SPECIFIC PYQ OWNERSHIP

All three cards are verified direct routes: 2019 landslide zonation, 2021 Himalayan-versus-Western-Ghats causes, and 2021 landslide causes/effects/National Landslide Risk Management Strategy. Avalanche and GLOF content is not misrepresented as printed in those stems.

PYQ DEMAND CARD 1 - 2019 GS-III

Demand: Explain landslide hazard zonation mapping and its role in preparedness and mitigation.

Status: Verified direct routing: Explain · 15 marks · 250 words; the answer distinguishes susceptibility, hazard and risk and converts mapping into decisions.

Model solution: Landslide: A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures. **Susceptibility versus hazard:** Susceptibility indicates where terrain is more likely to fail from relatively stable conditioning factors, whereas hazard adds probability, timing or intensity under triggers; neither alone equals risk. **Risk:** Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks. **Conditioning factors:** Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability. **GSI role:** GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work. **Zoning and land use:** Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan. **Slope and drainage measures:** Surface and subsurface drainage, controlled excavation, retaining or protective works, bioengineering and maintenance may reduce selected instability, but engineering must be site-specific and cannot guarantee every slope. **Warning and evacuation:** Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback. **Technology-outcome firewall:** A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

PYQ DEMAND CARD 2 - 2021 GS-I

Demand: Differentiate causes of landslides in the Himalayan region and Western Ghats.

Status: Verified direct routing: Differentiate · 10 marks · 150 words; use a comparison of conditioning factors, rainfall, seismicity and intervention without unsupported event statistics.

Model solution: Landslide: A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures. **Conditioning factors:** Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability. **Triggers:** Intense or prolonged rainfall, earthquakes, snow loading or warming, freeze-thaw, erosion, excavation and slope disturbance can trigger failure, but the relevant trigger differs among landslide, avalanche and GLOF mechanisms. **Cascading risk:** Earthquake or rain can trigger landslides that block rivers; lake or blockage failure can then generate flooding, while cloudburst, debris flow and slope failure may occur in one compound mountain emergency. **Zoning and land use:** Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan. **Slope and drainage measures:** Surface and subsurface drainage, controlled excavation, retaining or protective works, bioengineering and maintenance may reduce selected instability, but engineering must be site-specific and cannot guarantee every slope. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

PYQ DEMAND CARD 3 - 2021 GS-III

Demand: Describe landslide causes, effects and the National Landslide Risk Management Strategy.

Status: Verified direct routing: Describe · 15 marks · 250 words; strategy use remains bounded to mapping, prevention, warning, capacity and land-use functions.

Model solution: Landslide: A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures. **Risk:** Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks.

Conditioning factors: Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability. **Triggers:** Intense or prolonged rainfall, earthquakes, snow loading or warming, freeze-thaw, erosion, excavation and slope disturbance can trigger failure, but the relevant trigger differs among landslide, avalanche and GLOF mechanisms. **GSI role:** GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work. **Zoning and land use:** Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan. **Slope and drainage measures:** Surface and subsurface drainage, controlled excavation, retaining or protective works, bioengineering and maintenance may reduce selected instability, but engineering must be site-specific and cannot guarantee every slope. **Route and settlement planning:** Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services.

Warning and evacuation: Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback. **Institutional coordination:** GSI, DGRE, NDMA, CWC, NRSC/ISRO, State and district authorities, road agencies, hydropower operators and local communities hold different information and action roles that must be pre-agreed.

Technology-outcome firewall: A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

ORIGINAL MAINS 1 - 10 MARKS

Question: Distinguish landslide, rockfall, debris flow, avalanche and GLOF. Answer in about 150 words.

Model thesis: Claim: Landslide. **Named evidence/example:** A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures.

Analysis: This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Rockfall. **Named evidence/example:** Rockfall is the detachment and rapid fall, bounce or roll of rock from a steep face or slope, distinct from a coherent slide or channelised debris flow. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Debris flow. **Named evidence/example:** A debris flow is a rapid, channelised or slope-confined mixture of water, soil, rock and organic material; intense rainfall may trigger it but rainfall is not the moving mass. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Avalanche. **Named evidence/example:** An avalanche is rapid downslope movement of snow, ice and entrained material produced by snowpack instability, slope and weather conditions; it is not a landslide label for every mountain failure. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** GLOF. **Named evidence/example:** A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Claim -> named evidence -> analysis -> qualification:

- A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures.
- Rockfall is the detachment and rapid fall, bounce or roll of rock from a steep face or slope, distinct from a coherent slide or channelised debris flow.
- A debris flow is a rapid, channelised or slope-confined mixture of water, soil, rock and organic material; intense rainfall may trigger it but rainfall is not the moving mass.
- An avalanche is rapid downslope movement of snow, ice and entrained material produced by snowpack instability, slope and weather conditions; it is not a landslide label for every mountain failure.
- A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche.

Qualified conclusion: Claim: Landslide. **Named evidence/example:** A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Rockfall. **Named evidence/example:** Rockfall is the detachment and rapid fall, bounce or roll of rock from a steep face or slope, distinct from a coherent slide or channelised debris flow. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Debris flow. **Named evidence/example:** A debris flow is a rapid, channelised or slope-confined mixture of water, soil, rock and organic material; intense rainfall may trigger it but rainfall is not the moving mass. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Avalanche. **Named evidence/example:** An avalanche is rapid downslope movement of snow, ice and entrained material produced by snowpack instability, slope and weather

conditions; it is not a landslide label for every mountain failure. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** GLOF. **Named evidence/example:** A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

ORIGINAL MAINS 2 - 10 MARKS

Question: Differentiate landslide susceptibility, hazard and risk. Answer in about 150 words.

Model thesis: **Claim:** Susceptibility versus hazard. **Named evidence/example:** Susceptibility indicates where terrain is more likely to fail from relatively stable conditioning factors, whereas hazard adds probability, timing or intensity under triggers; neither alone equals risk. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Risk. **Named evidence/example:** Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** GSI role. **Named evidence/example:** GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhuvan material distinguishes mapping and experimental warning work. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Technology-outcome firewall. **Named evidence/example:** A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Claim -> named evidence -> analysis -> qualification:

- Susceptibility indicates where terrain is more likely to fail from relatively stable conditioning factors, whereas hazard adds probability, timing or intensity under triggers; neither alone equals risk.
- Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks.
- GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhuvan material distinguishes mapping and experimental warning work.
- A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence.

Qualified conclusion: **Claim:** Susceptibility versus hazard. **Named evidence/example:** Susceptibility indicates where terrain is more likely to fail from relatively stable conditioning factors, whereas hazard adds probability, timing or intensity under triggers; neither alone equals risk. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Risk. **Named evidence/example:** Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The

conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** GSI role. **Named evidence/example:** GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Technology-outcome firewall. **Named evidence/example:** A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

ORIGINAL MAINS 3 - 15 MARKS

Question: Analyse conditioning factors, triggers and cascading mountain risk. Answer in about 250 words.

Model thesis: **Claim:** Conditioning factors. **Named evidence/example:** Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Triggers. **Named evidence/example:** Intense or prolonged rainfall, earthquakes, snow loading or warming, freeze-thaw, erosion, excavation and slope disturbance can trigger failure, but the relevant trigger differs among landslide, avalanche and GLOF mechanisms. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Cascading risk. **Named evidence/example:** Earthquake or rain can trigger landslides that block rivers; lake or blockage failure can then generate flooding, while cloudburst, debris flow and slope failure may occur in one compound mountain emergency. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Route and settlement planning. **Named evidence/example:** Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Claim -> named evidence -> analysis -> qualification:

- Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability.
- Intense or prolonged rainfall, earthquakes, snow loading or warming, freeze-thaw, erosion, excavation and slope disturbance can trigger failure, but the relevant trigger differs among landslide, avalanche and GLOF mechanisms.
- Earthquake or rain can trigger landslides that block rivers; lake or blockage failure can then generate flooding, while cloudburst, debris flow and slope failure may occur in one compound mountain emergency.
- Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services.

Qualified conclusion: **Claim:** Conditioning factors. **Named evidence/example:** Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Triggers. **Named evidence/example:** Intense or prolonged rainfall, earthquakes, snow loading or warming, freeze-thaw, erosion, excavation and slope disturbance can trigger failure, but the relevant trigger differs among landslide, avalanche and GLOF mechanisms. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified

evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Cascading risk. **Named evidence/example:** Earthquake or rain can trigger landslides that block rivers; lake or blockage failure can then generate flooding, while cloudburst, debris flow and slope failure may occur in one compound mountain emergency. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Route and settlement planning. **Named evidence/example:** Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

ORIGINAL MAINS 4 - 15 MARKS

Question: Explain glacial-lake monitoring, GLOF warning and downstream evacuation. Answer in about 250 words.

Model thesis: **Claim:** GLOF. **Named evidence/example:** A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** NRSC-ISRO role. **Named evidence/example:** NRSC and ISRO provide Earth-observation inventories and mapping support for landslides and glacial or water-body monitoring; an image or atlas is decision support, not proof of safe slopes or warned communities. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Glacial-lake monitoring. **Named evidence/example:** Glacial-lake risk screening requires repeated remote sensing, lake and dam-character assessment, upstream slope or ice-change observation, field validation where feasible and downstream exposure mapping. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Warning and evacuation. **Named evidence/example:** Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Institutional coordination. **Named evidence/example:** GSI, DGRE, NDMA, CWC, NRSC/ISRO, State and district authorities, road agencies, hydropower operators and local communities hold different information and action roles that must be pre-agreed. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Technology-outcome firewall. **Named evidence/example:** A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Claim -> named evidence -> analysis -> qualification:

- A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche.
- NRSC and ISRO provide Earth-observation inventories and mapping support for landslides and glacial or water-body monitoring; an image or atlas is decision support, not proof of safe slopes or warned communities.
- Glacial-lake risk screening requires repeated remote sensing, lake and dam-character assessment, upstream slope or ice-change observation, field validation where feasible and downstream exposure mapping.

- Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback.
- GSI, DGRE, NDMA, CWC, NRSC/ISRO, State and district authorities, road agencies, hydropower operators and local communities hold different information and action roles that must be pre-agreed.
- A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence.

Qualified conclusion: Claim: GLOF. **Named evidence/example:** A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** NRSC-ISRO role. **Named evidence/example:** NRSC and ISRO provide Earth-observation inventories and mapping support for landslides and glacial or water-body monitoring; an image or atlas is decision support, not proof of safe slopes or warned communities. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Glacial-lake monitoring. **Named evidence/example:** Glacial-lake risk screening requires repeated remote sensing, lake and dam-character assessment, upstream slope or ice-change observation, field validation where feasible and downstream exposure mapping. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Warning and evacuation. **Named evidence/example:** Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Institutional coordination. **Named evidence/example:** GSI, DGRE, NDMA, CWC, NRSC/ISRO, State and district authorities, road agencies, hydropower operators and local communities hold different information and action roles that must be pre-agreed. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Technology-outcome firewall. **Named evidence/example:** A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

ORIGINAL MAINS 5 - 20 MARKS

Question: Evaluate zoning, drainage, slope treatment, road alignment and settlement planning for landslide risk. Answer in about 300 words.

Model thesis: Claim: Susceptibility versus hazard. **Named evidence/example:** Susceptibility indicates where terrain is more likely to fail from relatively stable conditioning factors, whereas hazard adds probability, timing or intensity under triggers; neither alone equals risk. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Risk. **Named evidence/example:** Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Conditioning factors. **Named evidence/example:** Geology, slope angle and shape, weathering, soil

or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** GSI role. **Named evidence/example:** GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Zoning and land use. **Named evidence/example:** Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Slope and drainage measures. **Named evidence/example:** Surface and subsurface drainage, controlled excavation, retaining or protective works, bioengineering and maintenance may reduce selected instability, but engineering must be site-specific and cannot guarantee every slope. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Route and settlement planning. **Named evidence/example:** Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Warning and evacuation. **Named evidence/example:** Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Technology-outcome firewall. **Named evidence/example:** A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Claim -> named evidence -> analysis -> qualification:

- Susceptibility indicates where terrain is more likely to fail from relatively stable conditioning factors, whereas hazard adds probability, timing or intensity under triggers; neither alone equals risk.
- Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks.
- Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability.
- GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work.
- Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan.
- Surface and subsurface drainage, controlled excavation, retaining or protective works, bioengineering and maintenance may reduce selected instability, but engineering must be site-specific and cannot guarantee every slope.
- Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services.

- Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback.
- A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence.

Qualified conclusion: Claim: Susceptibility versus hazard. **Named evidence/example:** Susceptibility indicates where terrain is more likely to fail from relatively stable conditioning factors, whereas hazard adds probability, timing or intensity under triggers; neither alone equals risk. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Risk. **Named evidence/example:** Risk combines the mass-movement or flood hazard with exposed people, routes, settlements, infrastructure and vulnerability or capacity, so a high-susceptibility unoccupied slope and a lower-susceptibility dense corridor can pose different risks. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Conditioning factors. **Named evidence/example:** Geology, slope angle and shape, weathering, soil or rock structure, drainage, land cover, toe erosion, prior movement and human cutting or loading condition slope stability. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** GSI role. **Named evidence/example:** GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Zoning and land use. **Named evidence/example:** Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Slope and drainage measures. **Named evidence/example:** Surface and subsurface drainage, controlled excavation, retaining or protective works, bioengineering and maintenance may reduce selected instability, but engineering must be site-specific and cannot guarantee every slope. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Route and settlement planning. **Named evidence/example:** Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Warning and evacuation. **Named evidence/example:** Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Technology-outcome firewall. **Named evidence/example:** A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

ORIGINAL MAINS 6 - 20 MARKS

Question: Design a coordinated Himalayan landslide-avalanche-GLOF governance framework under uncertainty. Answer in about 300 words.

Model thesis: **Claim:** Landslide. **Named evidence/example:** A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Avalanche. **Named evidence/example:** An avalanche is rapid downslope movement of snow, ice and entrained material produced by snowpack instability, slope and weather conditions; it is not a landslide label for every mountain failure. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** GLOF. **Named evidence/example:** A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Cascading risk. **Named evidence/example:** Earthquake or rain can trigger landslides that block rivers; lake or blockage failure can then generate flooding, while cloudburst, debris flow and slope failure may occur in one compound mountain emergency. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** GSI role. **Named evidence/example:** GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** DGRE role. **Named evidence/example:** DGRE under DRDO is the hazard-specific institutional anchor for snow and avalanche observation, assessment and warnings; its bulletin is not a GSI landslide forecast. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** NRSC-ISRO role. **Named evidence/example:** NRSC and ISRO provide Earth-observation inventories and mapping support for landslides and glacial or water-body monitoring; an image or atlas is decision support, not proof of safe slopes or warned communities. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Glacial-lake monitoring. **Named evidence/example:** Glacial-lake risk screening requires repeated remote sensing, lake and dam-character assessment, upstream slope or ice-change observation, field validation where feasible and downstream exposure mapping. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Zoning and land use. **Named evidence/example:** Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Route and settlement planning. **Named evidence/example:** Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Warning and evacuation. **Named evidence/example:** Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation

triggers, accessible communication, safe destinations, drills and feedback. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Institutional coordination. **Named evidence/example:** GSI, DGRE, NDMA, CWC, NRSC/ISRO, State and district authorities, road agencies, hydropower operators and local communities hold different information and action roles that must be pre-agreed. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Technology-outcome firewall. **Named evidence/example:** A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Claim -> named evidence -> analysis -> qualification:

- A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures.
- An avalanche is rapid downslope movement of snow, ice and entrained material produced by snowpack instability, slope and weather conditions; it is not a landslide label for every mountain failure.
- A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche.
- Earthquake or rain can trigger landslides that block rivers; lake or blockage failure can then generate flooding, while cloudburst, debris flow and slope failure may occur in one compound mountain emergency.
- GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work.
- DGRE under DRDO is the hazard-specific institutional anchor for snow and avalanche observation, assessment and warnings; its bulletin is not a GSI landslide forecast.
- NRSC and ISRO provide Earth-observation inventories and mapping support for landslides and glacial or water-body monitoring; an image or atlas is decision support, not proof of safe slopes or warned communities.
- Glacial-lake risk screening requires repeated remote sensing, lake and dam-character assessment, upstream slope or ice-change observation, field validation where feasible and downstream exposure mapping.
- Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan.
- Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services.
- Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback.
- GSI, DGRE, NDMA, CWC, NRSC/ISRO, State and district authorities, road agencies, hydropower operators and local communities hold different information and action roles that must be pre-agreed.
- A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence.

Qualified conclusion: **Claim:** Landslide. **Named evidence/example:** A landslide is gravity-driven downslope movement of rock, debris or earth; the movement class and material should be identified before assigning triggers or measures. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Avalanche. **Named evidence/example:** An avalanche is rapid downslope movement of snow, ice and entrained material produced by snowpack instability, slope and weather conditions; it is not a landslide label for every mountain failure. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** GLOF. **Named evidence/example:** A Glacial Lake Outburst Flood is the sudden release of impounded glacial-lake water after failure, overtopping or displacement affecting a natural dam, producing a downstream flood rather than a snow avalanche. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Cascading risk. **Named evidence/example:** Earthquake or rain can trigger landslides that block rivers; lake or blockage failure can then generate flooding, while cloudburst, debris flow and slope failure may occur in one compound mountain emergency. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** GSI role. **Named evidence/example:** GSI owns national landslide susceptibility, inventory, hazard/risk research and forecasting-development functions; its Bhusanket material distinguishes mapping and experimental warning work. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication.

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Qualification: The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** NRSC-ISRO role. **Named evidence/example:** NRSC and ISRO provide Earth-observation inventories and mapping support for landslides and glacial or water-body monitoring; an image or atlas is decision support, not proof of safe slopes or warned communities. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Glacial-lake monitoring. **Named evidence/example:** Glacial-lake risk screening requires repeated remote sensing, lake and dam-character assessment, upstream slope or ice-change observation, field validation where feasible and downstream exposure mapping. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Zoning and land use. **Named evidence/example:** Susceptibility or hazard zonation should guide avoidance, conditional development, route alignment, settlement expansion, drainage, site investigation and evacuation planning rather than merely decorate a plan. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Route and settlement planning. **Named evidence/example:** Mountain roads and settlements need risk-sensitive alignment and siting, spoil and drainage control, safe stoppage or closure protocols, alternative access, assembly areas and protection of critical services. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Warning and evacuation. **Named evidence/example:** Effective warning connects monitoring and forecast uncertainty to authorised messages, route closure or evacuation triggers, accessible communication, safe destinations, drills and feedback. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Institutional coordination. **Named evidence/example:** GSI, DGRE, NDMA, CWC, NRSC/ISRO, State and district authorities, road agencies, hydropower operators and local communities hold different information and action roles that must be pre-agreed. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Technology-outcome firewall. **Named evidence/example:** A susceptibility map, satellite image, sensor, bulletin, portal, protective structure or programme proves a capability or input; prediction skill, message receipt, evacuation, route safety and reduced loss require separate dated evidence.

Analysis: This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.